

2. 2.1 How many numbers are there between 25 and 250 that these numbers are divided by 7 ? (5 marks)

2.2 Given the finite geometric series such that $64 + a + b + c + \frac{1}{4}$, find the total sum of the series and determine the values of a , b and c . (5 marks)

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4 4.1 Show that the sequence $\left\{ \frac{n(4n+1)(5n+3)}{6n^2+2} \right\}_{n=1}^{\infty}$ converges. (3 marks)

4.2 Determine whether the sequence $\left\{ \sqrt{n}(\sqrt{n+1} - \sqrt{n}) \right\}_{n=1}^{\infty}$ converges. (4 marks)

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4.3 Determine whether the sequence $\left\{ \frac{7^{n+2} + 2^{n-1}}{7^{n+1} - 5^{n+1}} \right\}_{n=1}^{\infty}$ converges. (3 marks)

6. Suppose that $\vec{u} = 4\hat{i} - 3\hat{j} + 5\hat{k}$, $\vec{v} = -3\hat{i} - 5\hat{j} - 2\hat{k}$. Find

6.1 $\vec{w}$ satisfy condition $\vec{u} - 3\vec{v} + 2\vec{w} = \vec{0}$. (5 marks)

6.2 $(\vec{u} \cdot \vec{w})(\vec{u} \times \vec{v})$. (5 marks)

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8. 8.1 Given that $\vec{u}$, $\vec{v}$ and $\vec{w}$ are vectors in rectangular coordinate.

Let $\vec{u} = a\hat{i} + b\hat{j} + c\hat{k}$ when a, b, c are real number, $\vec{v} = \hat{i} + \hat{j} - \hat{k}$

and $\vec{w} = 2\hat{i} - \hat{j} + 2\hat{k}$. If $\vec{u}$ has same direction of $\vec{v} \times \vec{w}$ and $|\vec{u}| = 5\sqrt{26}$.

Find the value of $a - b + c$. (5 marks)

8.2 Compute the volume of the parallelepiped which generated by the vectors $\vec{u}, \vec{v}, \vec{w}$.

(5 marks)

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1. Determine the general term (a_n) of the following sequences

1.1 The sequence $\frac{1}{2}, \frac{3}{4}, \frac{7}{6}, \frac{9}{8}, \dots, a_n, \dots$ (3 marks)

1.2 The sequence $1, (1+2), (1+2+3), (1+2+3+4), \dots, a_n, \dots$ (3 marks)

1.3 The sequence $1, 7, 25, 79, 241, \dots, a_n, \dots$ (4 marks)

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3 3.1 The common ratio of a geometric series is $r = -\frac{2}{3}$. The total sum of this series is 13 with $a_n = -8$. How many terms are there in this series? (5 marks)

3.2 The arithmetic series satisfies the expressions $a_3 + a_7 = 13$ and $a_1 + a_2 + a_3 + a_4 + \dots + a_{13} = 104$. Evaluate the values of the first three terms a_1, a_2, a_3 . (5 marks)

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5. Given that $A = 4 + \frac{4}{1+2} + \frac{4}{1+2+3} + \frac{4}{1+2+3+4} + \dots$

and $B = 1 + \frac{3}{4} + \frac{7}{16} + \frac{13}{64} + \frac{21}{256} + \dots$. Find the value of $27A + B$. (10 marks)

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7. 7.1 Let $A(-1, -2, -2)$, $B(0, -1, 0)$ and $C(0, 0, c)$, when $c > 0$ be the vertices of a triangle. If the area of the triangle is $\frac{\sqrt{11}}{2} \text{ unit}^2$, find c^2 . (5 marks)

7.2 Assume that $|\vec{u} + \vec{v}| = \sqrt{37}$, $|\vec{u} - \vec{v}| = \sqrt{13}$ and the angle between the vector $\vec{u}$ and the vector $\vec{v}$ is 60° . Determine the value of $|\vec{u}| + |\vec{v}|$. (5 marks)

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